UNIVERSITI TEKNOLOGI MARA

ICT602
MOBILE TECHNOLOGY AND DEVELOPMENT

TOPIC 2 – Mobile Devices

LESSON OUTCOME

Identify, explain and Discuss:

1. Device taxonomy
2. General purpose devices
3. Ubiquitous computing
4. Location-Awareness and Context-Awareness

DEVICE TAXONOMY

Mobile Device

- a **portable computing device** such as a smartphone or tablet computer.
- "Mobile device" is a general term for any handheld computer or smartphone.
- A mobile device is a handheld tablet or other device that is made for portability, and is therefore both compact and lightweight.

Device Taxonomy

- Refers to the **classification and categorization of mobile devices** based on specific criteria such as hardware features, intended usage or user preference.
  - **Device Type -** Smartphones, tablets, phablets, smartwatches, etc
  - **Performance and Features -** Devices can be classified based on hardware performance (e.g., high-end flagship devices vs. budget phones) and key features (e.g., large display, camera quality, battery life, processing power).
  - **The intended use of the device / use case -** Communication (e.g., voice-centric devices), productivity (e.g., larger screen tablets for professionals), multimedia consumption, gaming, or social networking.

Device Taxonomy: Users' Device of Choice

- **An email centric user** might want a RIM Blackberry or Palm Treo, devices that have rich information service but less than ideal voice experiences.
- Another user who is a **salesperson** may prefer voice phone and would put high priority on voice-based communication than data services.
- An internet marketer or blogger might need to see large amounts of information simultaneously may prefer device with a large screen (Phablet or Tablet)
- A student immersed in social networking would like a device focused on instant messaging and the web.

Device Taxonomy: Users' Device of Choice

- For these reasons, mobile computing devices **will not converge on a single physical form**.
- Instead devices will always be designed to **fit the needs** of users based on classes.

Device Taxonomy: Users' Device of Choice

- A typical mobile user has several characteristic, they are usually: Interruptible, Easily distracted, Available, Sociability, and Identifiable.
- However, regardless of these common characteristics, each **users needs and interests varies greatly**.
- Users has different needs and interest although they share common characteristics. These **interests and needs affect users' choices** in devices.

Device Taxonomy: Users' Device of Choice

- **Salesperson**
  Prefers mobile phone with strong support for calls, messaging, and calendar/scheduling apps.

  Needs fast access to CRM apps, navigation, and voice assistants like Siri or Google Assistant.

Device Taxonomy: Users' Device of Choice

- **Digital Marketer / Content Creator**
  Prefers mobile phone or tablet (like a Galaxy Tab or iPad Pro) for better viewing and editing.

  Needs smooth use of video, photo editing, social media apps, and cloud sharing.

  Emphasis on devices that support multi-tasking and stylus input.

Device Taxonomy: Users' Device of Choice

- **Student / Youth User**
  Uses a mid-range smartphone focused on social media, instant messaging, video calls, and web browsing.

  Needs reliable support for e-learning apps, YouTube, Google Classroom.

  Prefers devices with good battery life and camera quality.

Device Taxonomy: Users' Device of Choice

- **Gamers and Entertainment Seekers**
- Lean towards gaming phones, 5G tablets, or dedicated handheld consoles with high refresh rates (90Hz+), haptic feedback, and immersive graphics.

GENERAL PURPOSE & TARGETED DEVICES

Mobile Devices

- Mobile Devices can be categorized into two categories.
  - ***General Purpose Devices***
  - ***Targeted Devices***

General Purpose Device

- General Purpose Devices are intended to target specific market needs. These devices are likely to be used frequently within their domain
  - **Entry level Device**
  - **Mid Range Device**
  - **Flagship**
  - **Premium**

General Purpose Device

- Most modern workers use computing device while working
- Although PCs are still prevalent in workplaces, some specialize work nowadays depends on Mobile Devices.

General Purpose Device

- multi-purpose devices with an entertainment focus, likely to be **near the user when entertainment is acceptable.**
- General Purpose Devices have **entertainment application**
- While entertainment device might be focused, it will still have **add-on features.**

General Purpose Device: Entertainment

- Keeping up with General Purpose Device with Entertainment as its <u>**Primary function**</u> may have <u>**add-on features**</u>.
- For example a **device focused on music and video** may have book reader, text messenger or a GPS map as an <u>**add-on feature.**</u>
- A device which **focus on Gaming** as its <u>primary feature</u> may be distinguished by having obvious industrial design that highlight gaming controls such as **Sony Xperia Play.**

General Purpose Device: Primary vs Addon feature

Samsung Galaxy Zoom – a successor to **Samsung Galaxy Camera** – which combines camera functionality as primary feature and internet application and phone as **<u>add-on feature</u>.**

General Purpose Device: Primary vs Addon feature

- A device with **primary feature** is usually designed to appeal other market segments while still offering **other features as add-on**
- Device that targets certain feature **may sacrifice usability** and ease of use on other features/functionality
- QUESTION: How to differentiate between **primary feature** and **addon-feature?**

Targeted Device

- Targeted device is also known as '**information appliance**'

- Targeted devices are designed to **help the user do a small number of tasks, and to do them well**

- Because of it's simplicity, it cannot provide other facility to support other non-trivial operations

Targeted Device

Example Targeted Device

- A music device needs data to play and would not perform other task besides playing music or connect to the internet to play music : iPod, Microsoft Zune

- A handheld digital camera :Samsung Galaxy NX, Nikon Coolpix S3000-series

- A handheld product price finder with barcode reader is only usable for its intended purpose of checking price leaving little room for other functionality.

Targeted Device use case

- **Retail** - Scanner that tracks inventory
- **Transportation and Logistics** – asset tracking and management
- **Hospitality** – Kiosks for food ordering, info or payment
- **Healthcare** – Monitor patients remotely, keep track of patient records from the cloud

https://www.android.com/intl/en_in/enterprise/devices/

ENTERPRISE ANDROID FOR HEALTHCARE

UBIQUITOUS COMPUTING

Ubiquitous

Anytime, anywhere, ….. Anyone, any device, any platform, … anything

Ubiquitous Computing

- Computers everywhere – computing is embedded everywhere
- Ubicomp is the method of enhancing computer use by making it **AVAILABLE throughout the physical** environment while **effectively INVISIBLE** to users.
- The idea of ubicomp arose from contemplating the place of today's computer in actual activities of everyday life
- The challenge is to create **a new kind of relationship of people to computers** - the 'presence' of computer would not get in the way of people lives.

Ubiquitous computing

- **Ubicomp helped kicked-off mobile computing** research interests around the world.

- Modern mobile devices has elements of ubiquitous computing where most users **does not realize** or feels like they are interfacing with computers while using mobile devices
  - Machine often can **'sense' changes** in environment and acts accordingly.

Ubiquitous Computing Principles

1) Help user perform activities in their daily life
2) Information and tasks should be made available everywhere
3) Computer should be '**invisible**' to users
4) Technology should **create calm** – in face of information overload, information representation should be well integrated within physical environment and should be intuitive to users

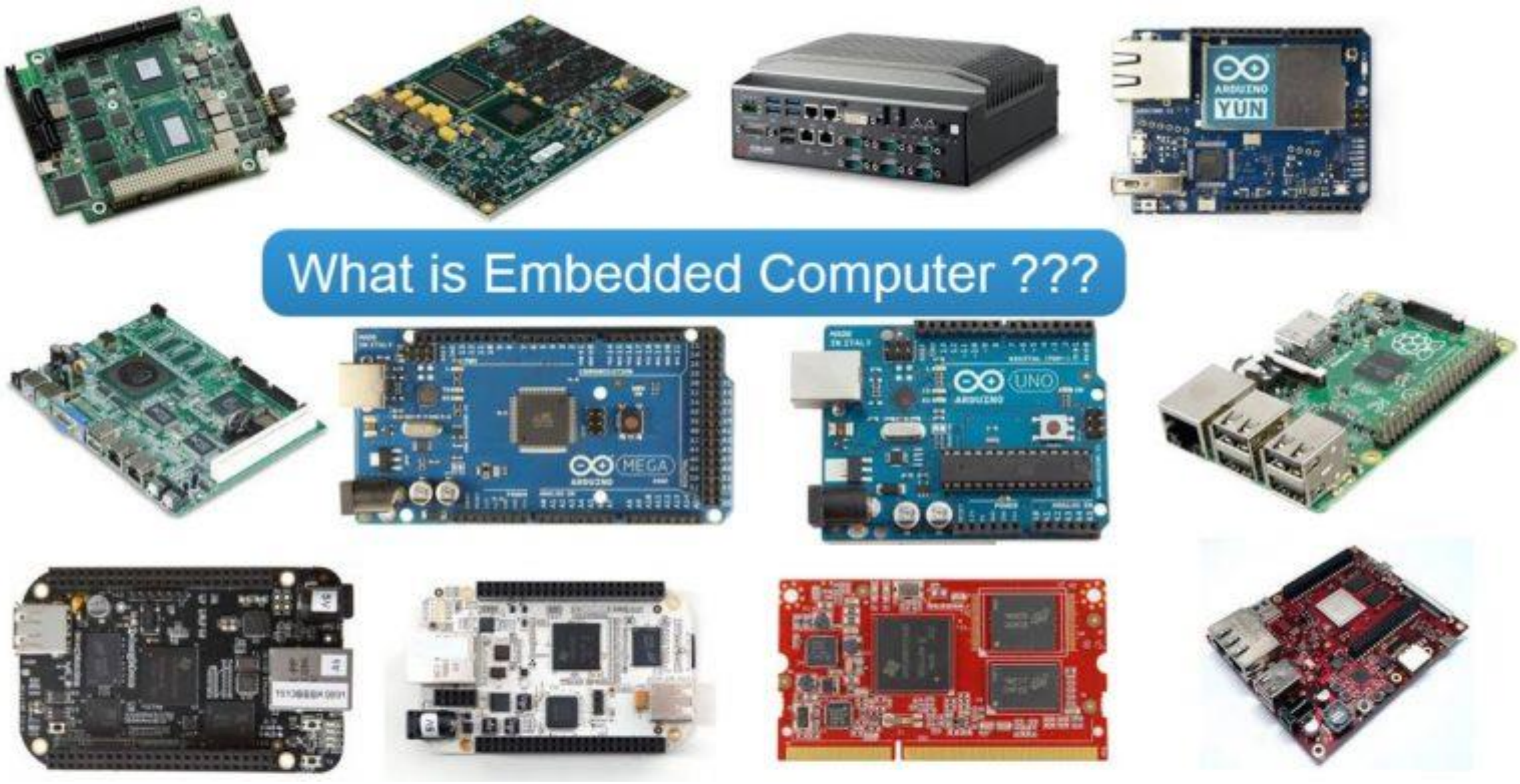
Ubiquitous Computing

Ubiquitous computing **vision**:

1) Desktop computers should be replaced by **embedded computers** in typical physical objects, without interfering with the current functionality of the said objects.

2) The use of embedded computers in typical object may **enhance object functionality**.

Embedded Computers



Embedded Computers

- An **Embedded Computer** is a **microcontroller** or **microprocessor-**based system devised for specific function.
- performs specific functionality and used today in various applications.
- may be a part of large system but it relies on its **own processor.**

General Computer vs Embedded Computer

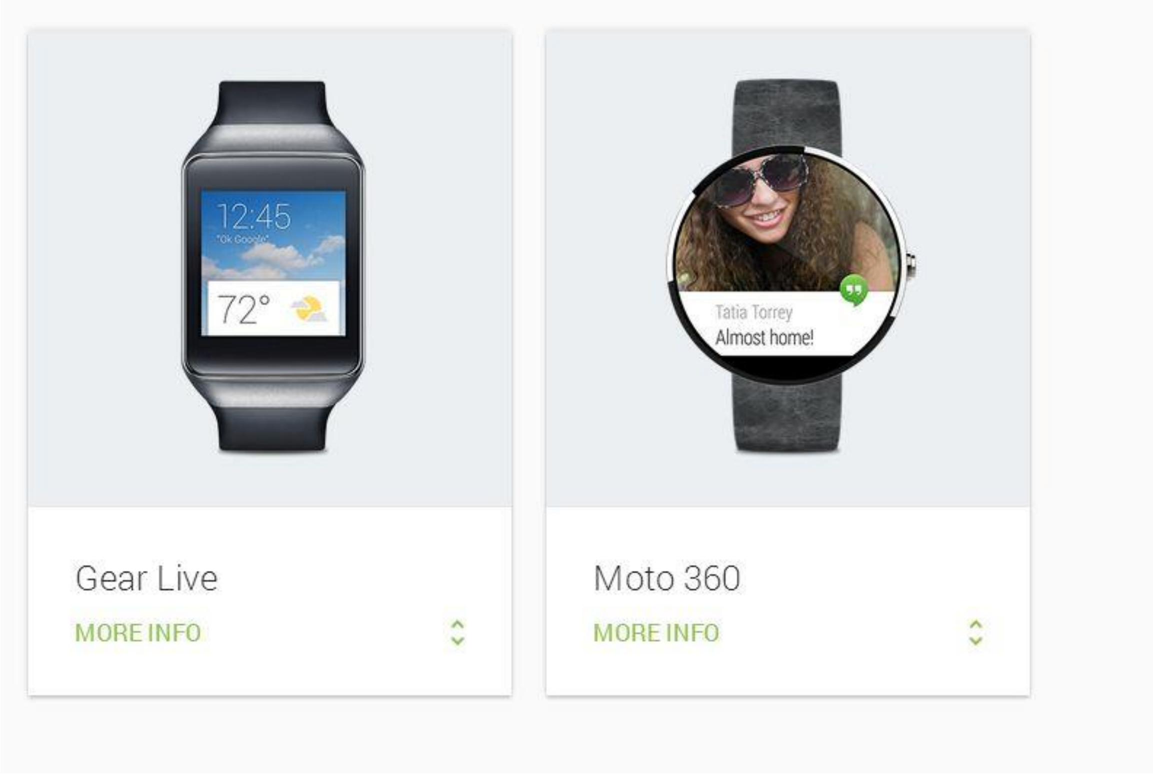
- A general-purpose computer are used for **different tasks**.
- Embedded Computer can only work on a **specific task.**
- A PC can work on **different functions at the same time**.
- General computers can work at different tasks while running different application software.
- Embedded computer are programmed for **functionality.**
- Embedded computer **does not require much power**

Example of Ubiquitous Computing Devices

- Smart phone
- Smart Watch
- Interactive White Board
- RFID Smart Tags
- Digital Cameras
- Anti-Glare Car Rear-View Mirror
- Google Glass
- Android Wear / Gear Live / Samsung Fit Gear watch

Ubicomp Device Examples

Wrist Watch



Ubicomp Device Examples

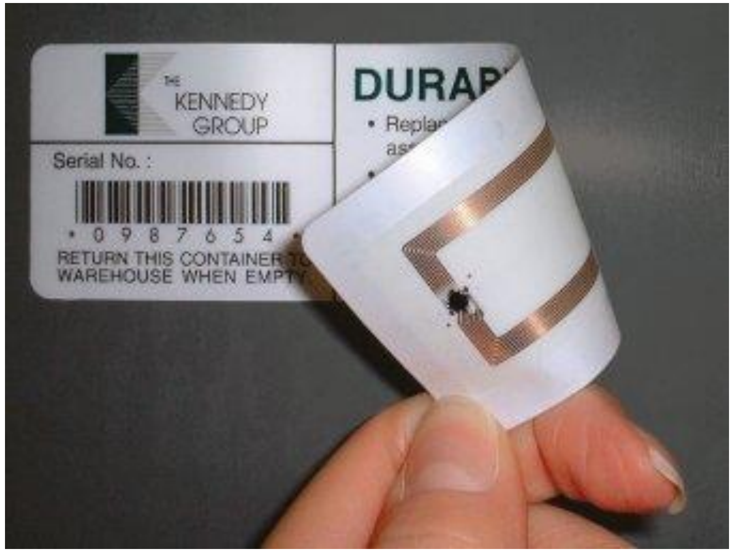
Camera

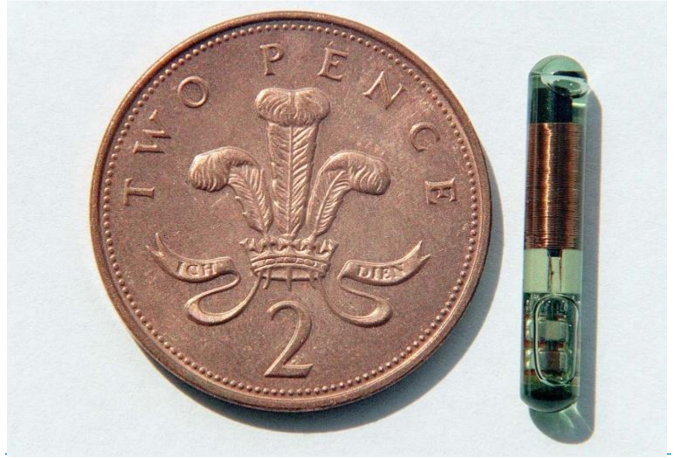
Ubicomp Device Examples

In-car Rear-view mirror

RFID Tags


gardeur
gardeur
RF-iT


THE KENNEDY GROUP
Serial No. :
0 9 8 7 6 5 4
RETURN THIS CONTAINER
WAREHOUSE WHEN EMPTY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Ubicomp: Three Waves of Computing

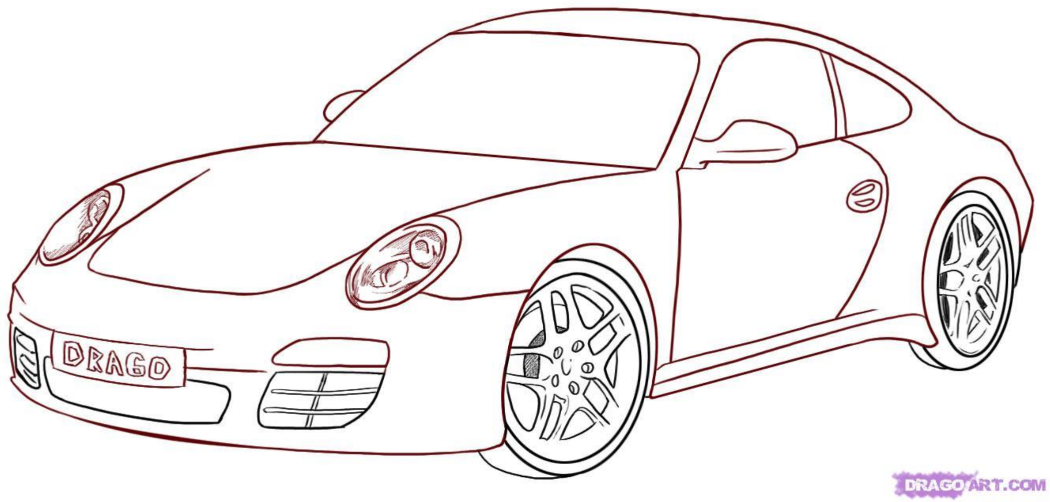
Computing has evolved in many ways, in essence, there are THREE(3) waves of computing:

1. **Main Frame:** Many person, One Computer

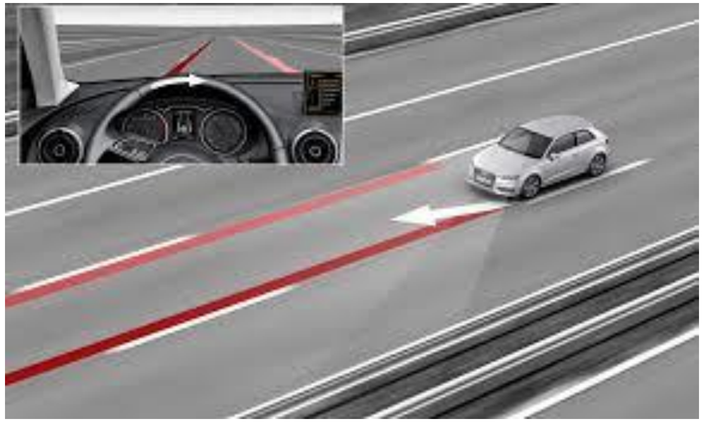
2. **Personal Computer (PC):** One Person, One Computer
3. **<u>Ubiquitous Computing</u>:** One person, Many Computers

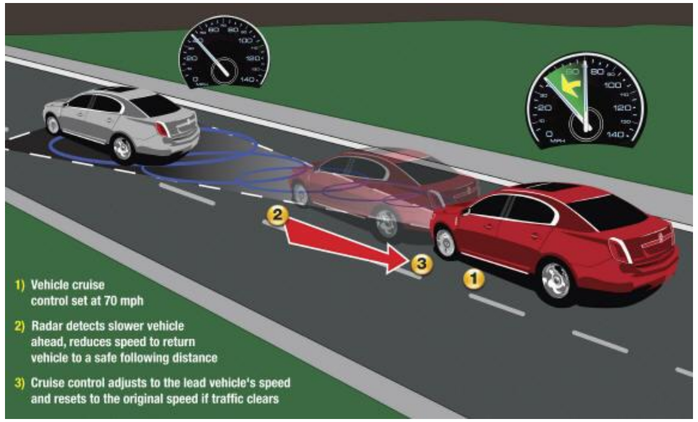
How many computers/sensors in a Car?

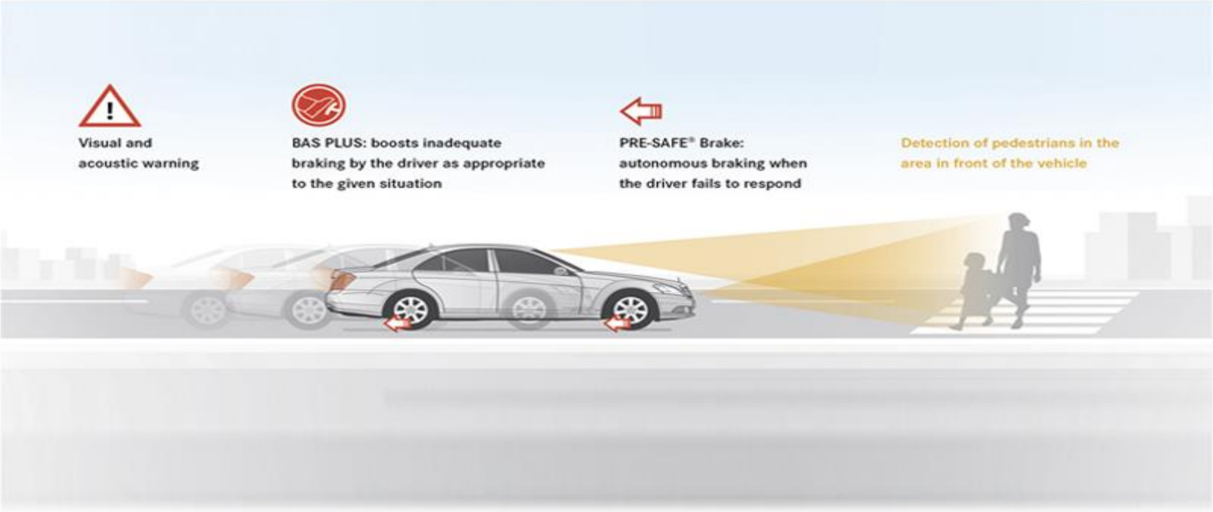
- **ECU – Engine Control Unit** :
  - Control ignition timing and idle speed.
  - Translates foot pedal pressure applied by a human to computer signals.
- **Transmission Computer (Transmission Control Module)** – which control transmission and gear ratio, especially in CVT-geared car
- **Driver Assistance System**
  - Lane Assist System
  - Blind Spot detection system
  - Park assist System
  - Adaptive cruise control system
  - Active braking system



How many computers/sensors in a Car?

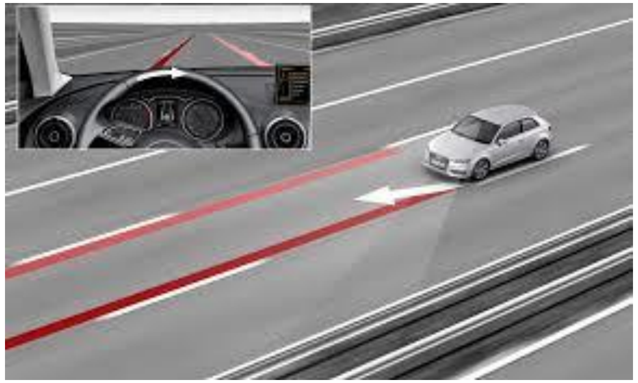


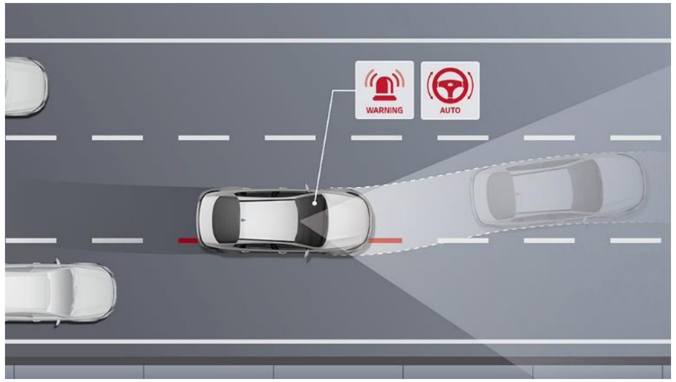


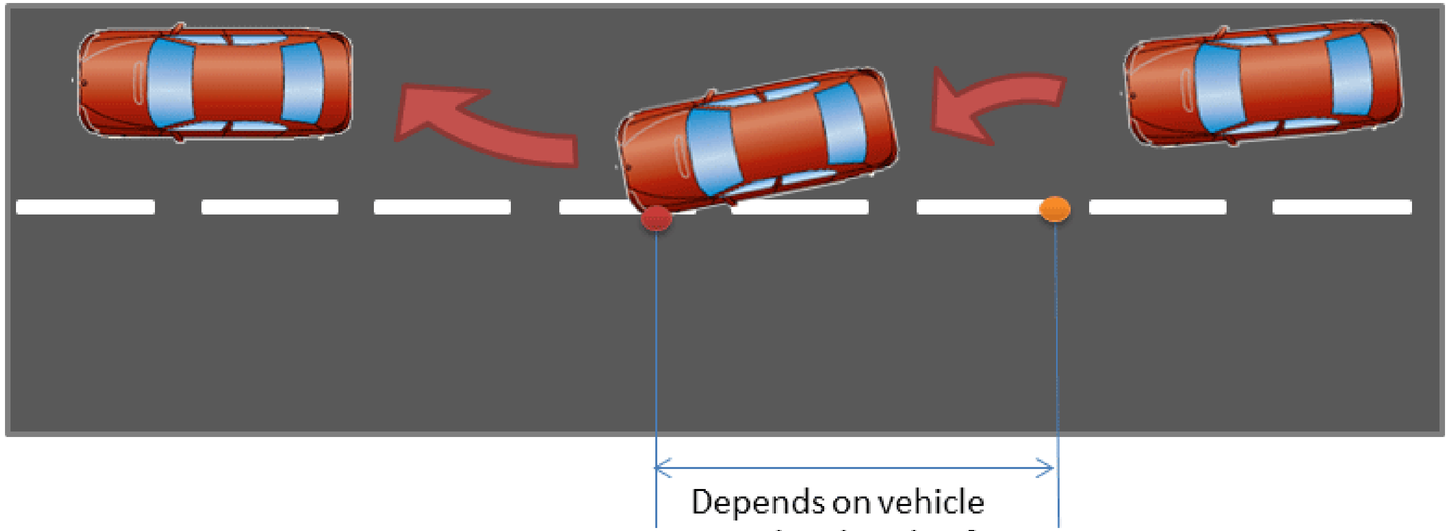


Lane Assist System

- https://www.youtube.com/watch?v=fLvJ_4ehPTk

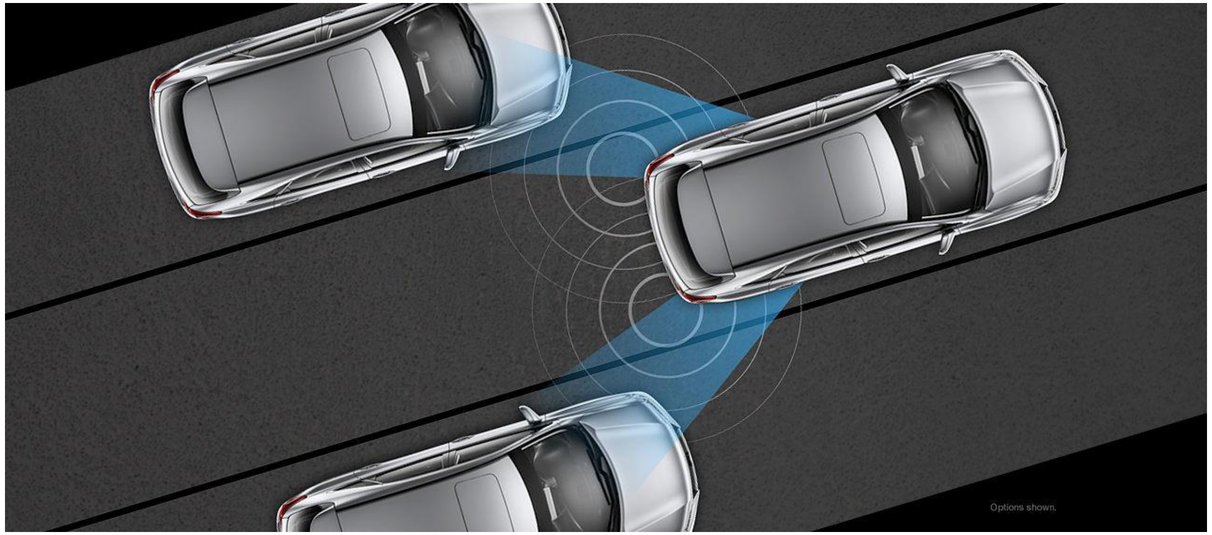


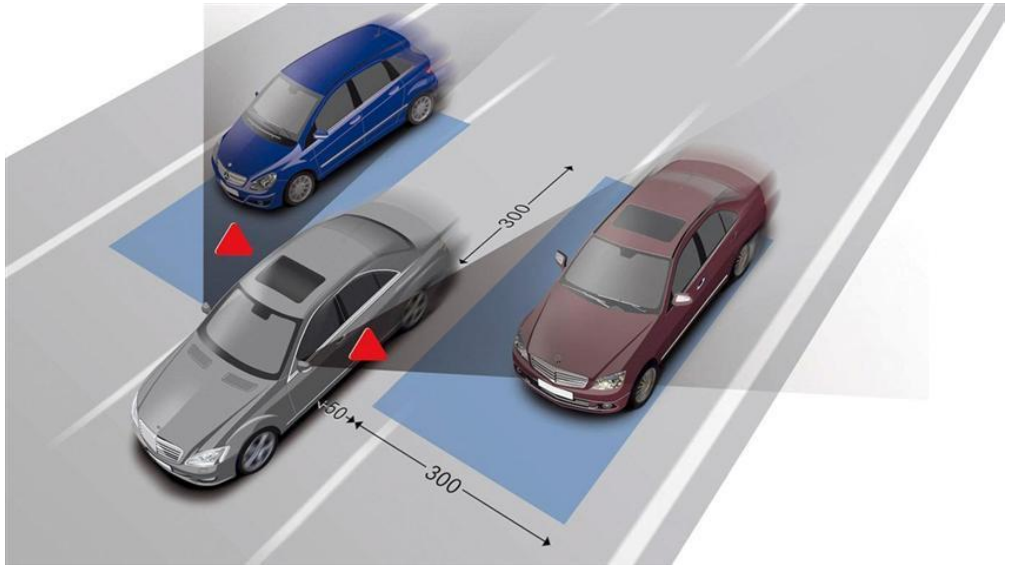




Blind Spot detection system

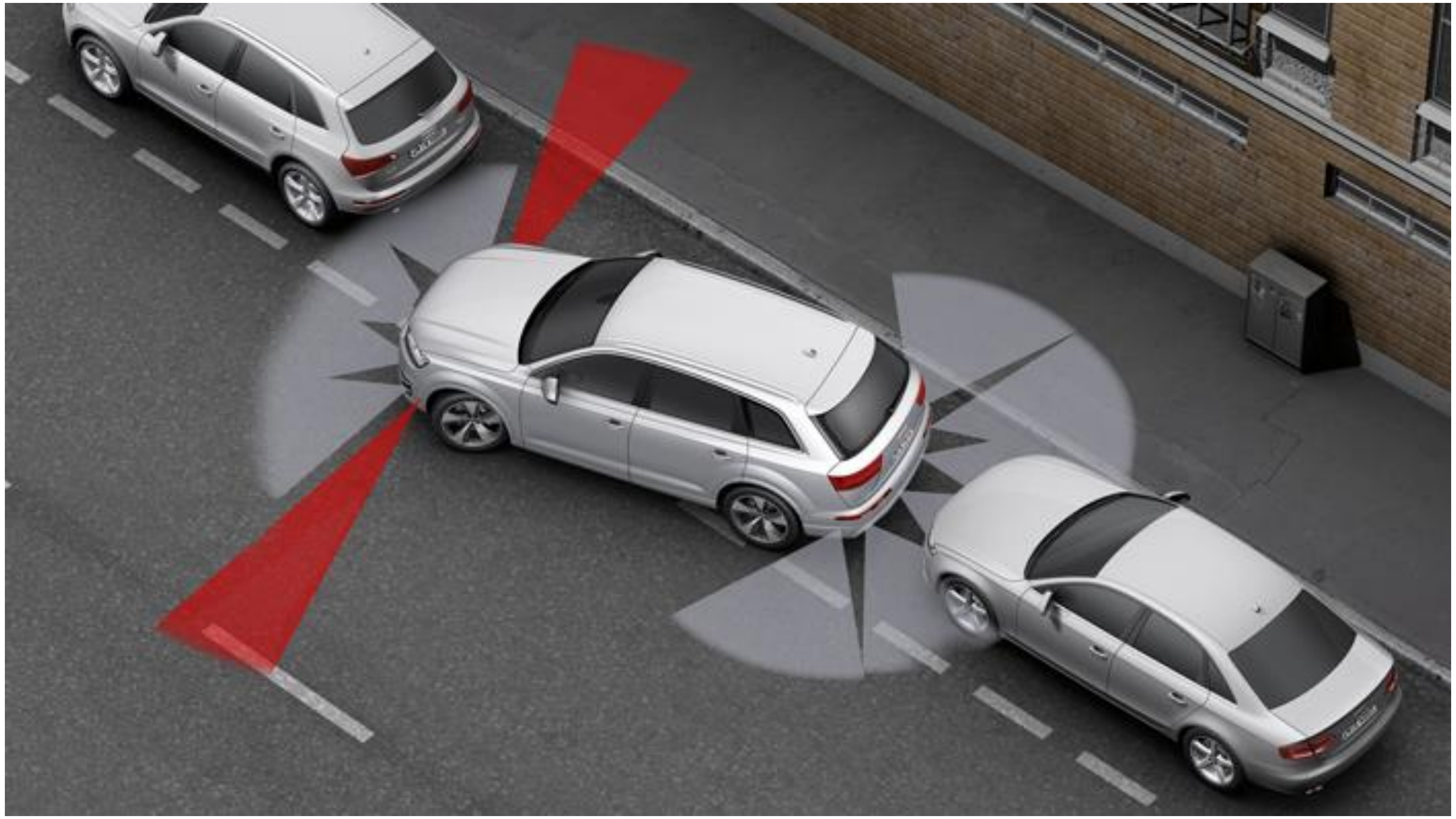
- https://www.youtube.com/watch?v=B93tfG4ZydY





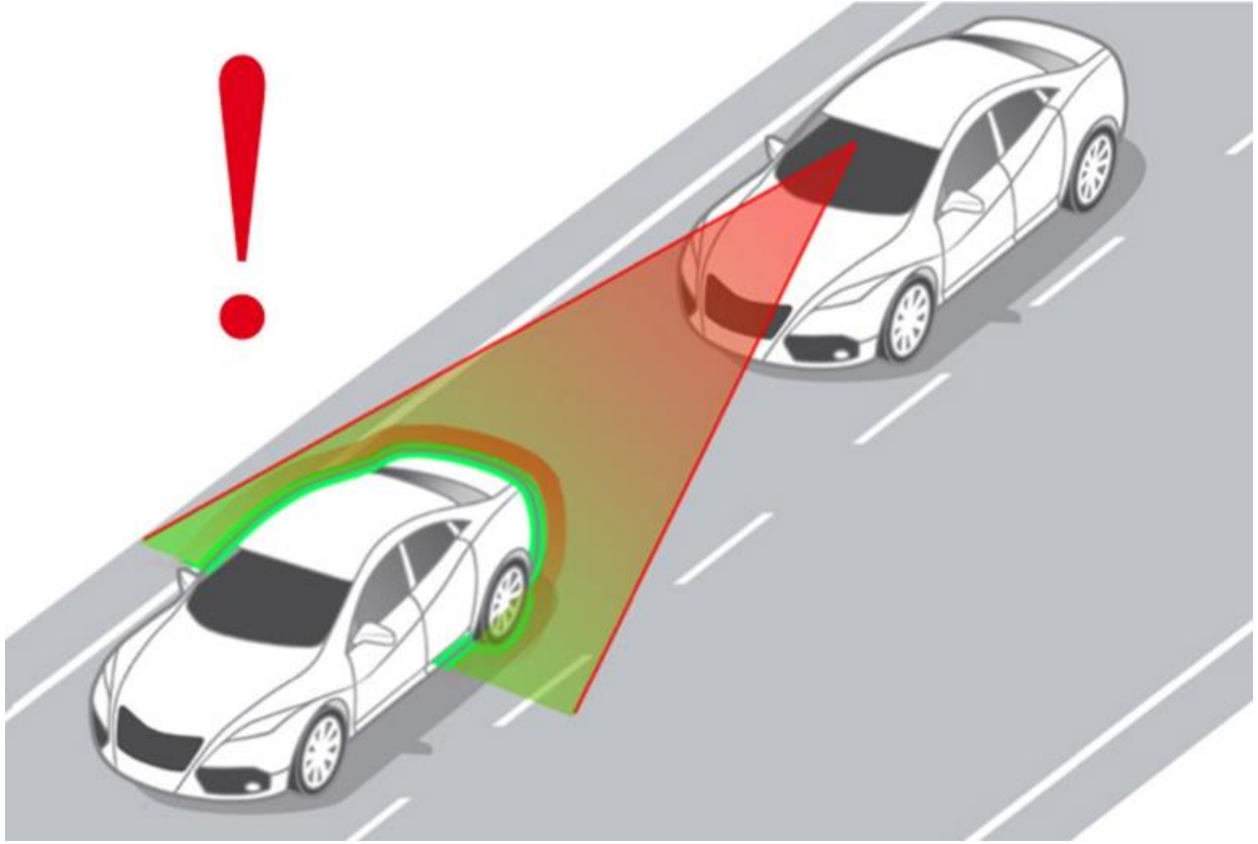
Park assist System

- https://www.youtube.com/watch?v=0Tg3g-HeWAQ



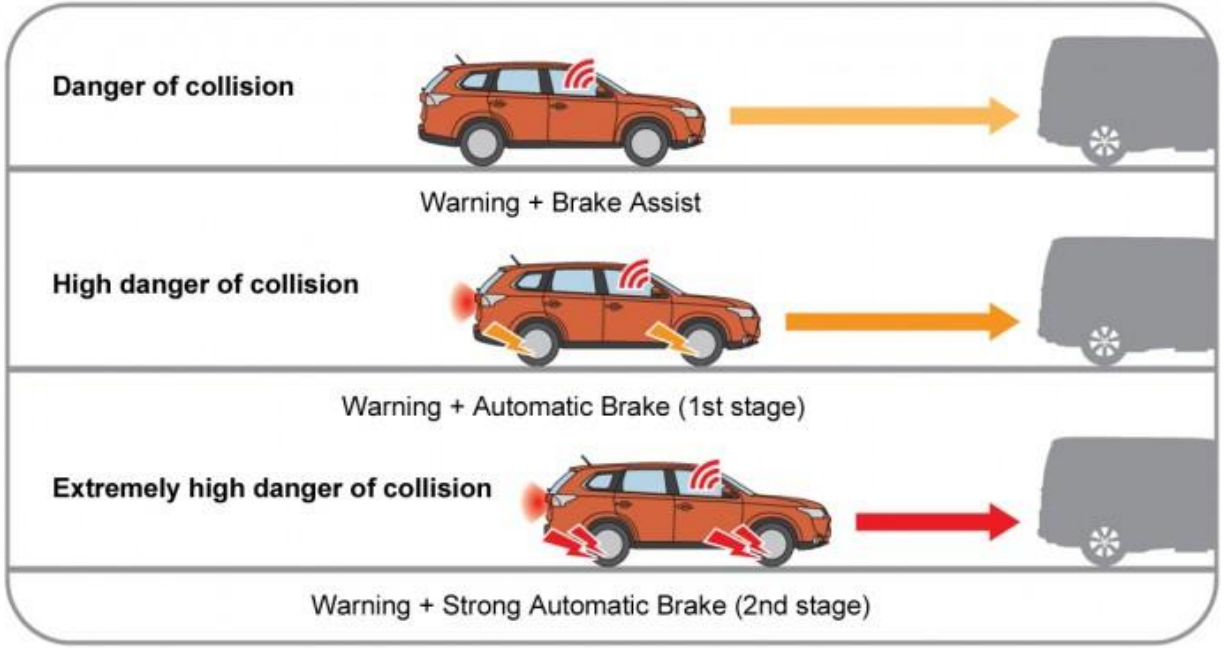
Adaptive cruise control system

- https://www.youtube.com/watch?v=GlnSPWZRFRM



Active braking system

- https://www.youtube.com/watch?v=ridS396W2BY
- https://www.youtube.com/watch?v=AQLOY-2GES4



Ubiquitous Computing Concept in Mobile Device

- **Always Available** - continuously connected, real-time comm.
- **Invisibility** – seamless integration
- **Context-Aware Interaction** – sensors to understand and react based on user context
- **Cloud Integration and Mobile Synchronization -** ubiquitous access to information across multiple devices

How many sensors in a smart phone

- Accelerometer
- Magnetometer
- Gyroscope (angular change sensor)
- Near Field Communication
- GPS + Assisted GPS
- Proximity sensor
- Photometer

References to Android CDD http://source.**android**.com/**compatibility**/**android**-cdd.pdf

Android CDD is the main reference for Device Manufacturers who wishes to receive "official Android" certification from Google.

Did you know: Phone sensors that may already assist you ?

- **Photometer** - adjust screen brightness automatically, thus save battery power
- **Accelerometer -** Automatic screen orientation
- **Proximity** - disable touch screen phone keypad while talking on the phone.

Key Elements of Ubicomp

- **Ubiquitous Networking** – as a medium to share data
- **Ubiquitous Sensing** – various senses/sensors which can act as ‘eyes and ears’ interacting with the environment to add ‘awareness’
- **Ubiquitous Access –** concept of accessing service/data from anywhere with natural, unobtrusive display, user require least effort to absorb information
- **Ubiquitous Middleware** – shield application from low level implementation details.

Examples of Devices/Applications implementing Key Elements of Ubiquitous computing

Google Fit / Apple Health

- **Ubiquitous Sensing:** Monitors steps, heart rate, workouts, sleep, and environmental movement.
- **Ubiquitous Networking:** Syncs with cloud profiles, other apps (e.g., Strava, MyFitnessPal).
- **Ubiquitous Access:** Works across phone, smartwatch, and fitness trackers; passive tracking.
- **Ubiquitous Middleware:** Aggregates data from sensors and third-party apps to present clear health metrics.

Examples of Devices/Applications implementing Key Elements of Ubiquitous computing

E-hailing (Uber / Grab / Gojek)

- **Ubiquitous Sensing:** Uses GPS, accelerometer, and compass to locate user/driver and detect ride activities.
- **Ubiquitous Networking:** Real-time location of drivers, ride-matching, and digital payment.
- **Ubiquitous Access:** Works across phone, smartwatch. The drivers and passengers can share their ride details through cloud sharing platforms.
- **Ubiquitous Middleware:** Aggregates data from sensors and ride suggestion based on behavior.

Examples of Devices/Applications implementing Key Elements of Ubiquitous computing

M-commerce (Amazon Shopping / Shopee / Lazada**)**

- **Ubiquitous Networking:** Live product inventory, shipment tracking, chat with sellers.
- **Ubiquitous Access:** Available via web, PWA, and native apps; voice shopping with Alexa (Amazon).
- **Ubiquitous Middleware:** Personalized recommendations, smart notifications, and simple UI for transactions.

Examples of Devices/Applications implementing Key Elements of Ubiquitous computing

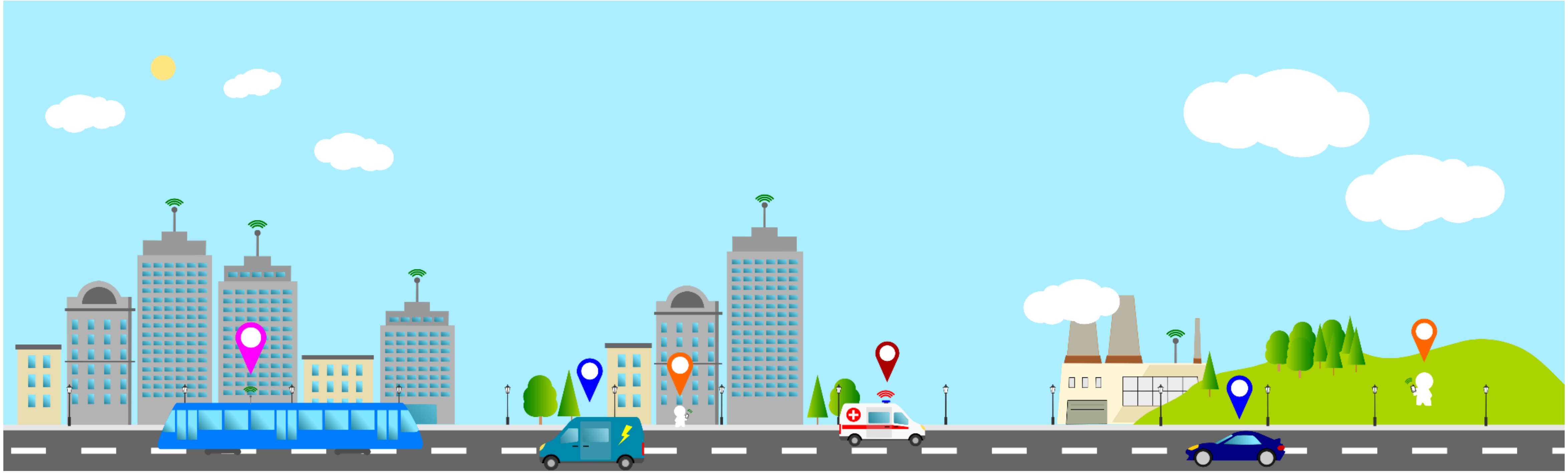
Camera/Gallery Application

- **Ubiquitous Access:** Synchronizing the photos/videos across multiple device with Google Photos or iCloud
- **Ubiquitous Middleware:** Face recognition, object detection, and event clustering without user intervention.
- **Ubiquitous Sensing:** Utilizes camera sensor, timestamp, location (GPS), and motion data for geotagging photos, detecting motion, compensate for sudden movements.

These systems function transparently, integrating into the user's environment without drawing attention.

LOCATION AWARENESS & CONTEXT AWARENESS

Location Awareness?



Location Awareness

Refers to ability of device or application to **provide VALUE-ADDED SERVICE based on user location.**

Information or services that can be relayed:

1) Local news and weather reports
2) Promotions and Advertisements
3) Navigation
4) Location Service
5) Exhibition


Explore
"http://m.flickr.com" Would Like To Use Your Current Location
Help / FAQ
Don't Allow
OK
A Yahoo! Company | Your Privacy | Terms of Service
flickr.com | old m.flickr.com | sign out
Explore
Upload
"Safari" Would Like to Use Your Current Location
Help / FAQ
Don't Allow
OK
A Yahoo! Company | Your Privacy | Terms of Service
flickr.com | old m.flickr.com | sign out
flickr
Signed in as joshlowensohn
Activity
You
Contacts
More
Photos taken nearby
French Quarter, San Francisco, CA, US
Google
Map data ©2009 Tele Atlas


New York
5:34 PM EDT
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New York
5:34 PM EDT
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Forecast
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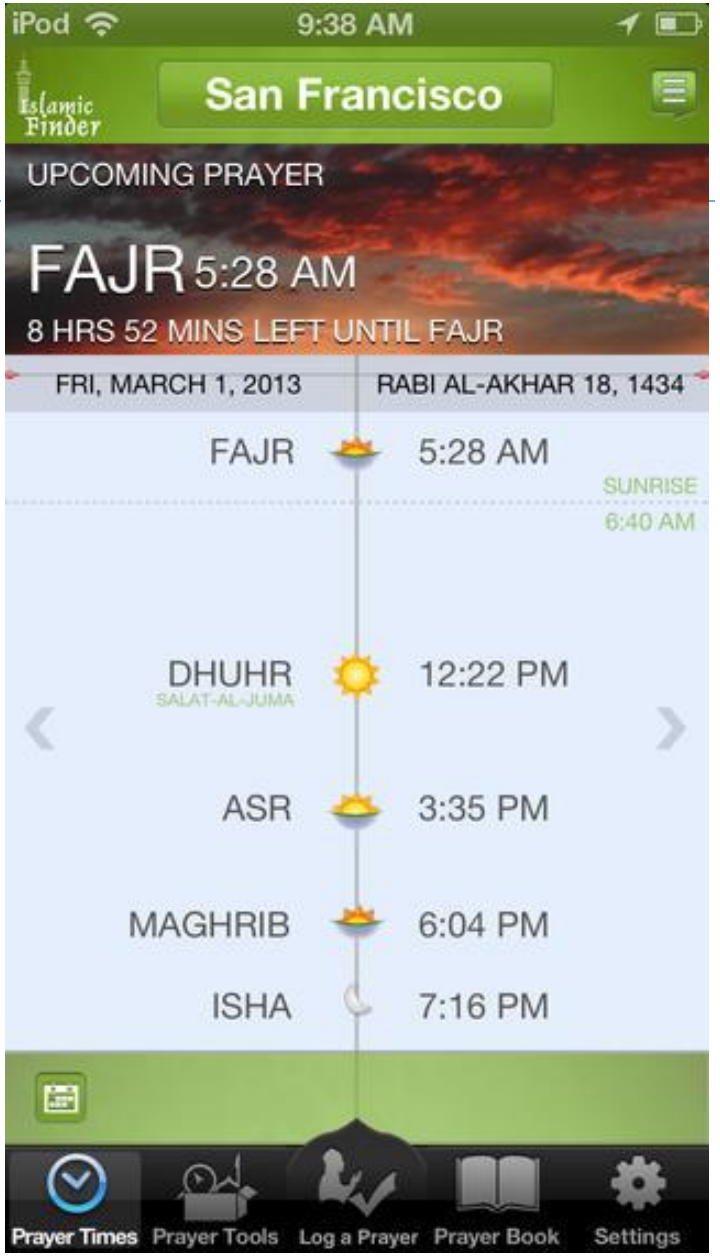
Sunday 57° 45°
Monday 57° 46°
Tuesday 64° 57°
Wednesday 70° 54°
Thursday 70° 57°


iPod 9:38 AM
Islamic Finder
San Francisco
UPCOMING PRAYER
FAJR 5:28 AM
8 HRS 52 MINS LEFT UNTIL FAJR
FRI, MARCH 1, 2013
RABI AL-AKHAR 18, 1434
FAJR 5:28 AM
SUNRISE 6:40 AM
DHUHR 12:22 PM
SALAT-AL-JUMA
ASR 3:35 PM
MAGHRIB 6:04 PM
ISHA 7:16 PM
Prayer Times
Prayer Tools
Log a Prayer
Prayer Book
Settings

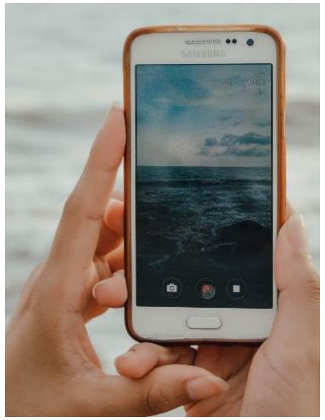
Advantages of Location Aware

1. Application can **better adapt to its environment**
2. Provide **useful information** that are relevant to the current user's situation
3. Helps user deals with information overload, by presenting information on subject nearest to users

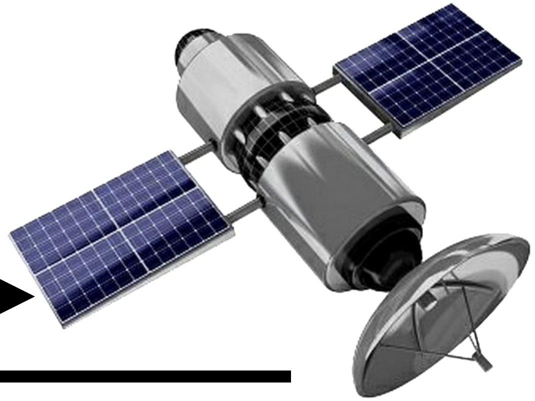
Application Areas

1) **Safety** - search and rescue and safety. User location can be pinpointed
2) **Social Networking** - enables users to share their updates and reviews to those who are nearby.
3) **Information Service & Business Exposure** - expose businesses to nearby users. Traffic accident warnings or advertisement
4) **Education** - Enables user to interact with their environment. Educational information regarding the surrounding area
5) **Tracking** - Tracking objects like pets, children or valuable items. navigation tool in outdoor activities like hiking and sailing

How Location Awareness implemented in Android Application?



1. User access a location aware application (eg: request a list of nearest pizza restaurant)

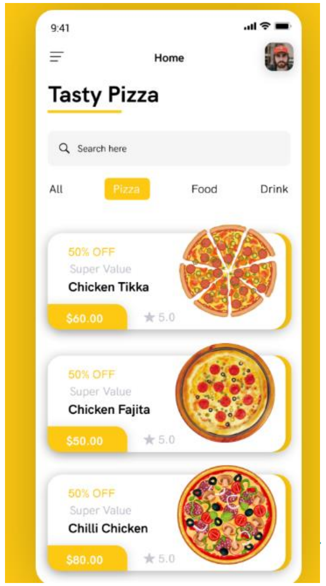


2. Mobile device gets estimated coordinates from Location Provider (GPS, WiFi or Assisted GPS)

3. The application makes requests to remote database with estimated GPS Coordinates

ID	Name	Lat	Lng	Address	Industry
0	Johns Pizzeria	40.7582	-73.9884	260 W 44th Street	Pizza
1	Lasso	40.721	-73.9954	192 Mott Street	Pizza
2	Lombardis Pizza	40.7215	-73.9956	32 Spring Street	Pizza
3	Adriennes Pizzabar	40.7043	-74.0102	54 Stone Street	Pizza
4	Solo Pizza NYC	40.7223	-73.9829	27 Avenue B	Pizza
5	Brick NYC	40.7144	-74.0077	22 Warren Street	Pizza
6	Luzzos	40.7656	-73.9579	211 1st Ave	Pizza
7	Lazzaras Pizza Cafe	40.7541	-73.9895	221 W 38th Street	Pizza
8	Johns of Bleecker Street	40.7317	-74.0034	278 Bleecker Street	Pizza
9	Co.	40.7472	-74.0008	230 9th Ave	Pizza

4. The remote server returns with relevant information based on the nearest user-location coordinates.



Examples of Location Aware Application

- EventBrite & Eventful
  - Discover popular local event and get event recommendation for the users

https://play.google.com/store/apps/details?id=com.eventbrite.attendee

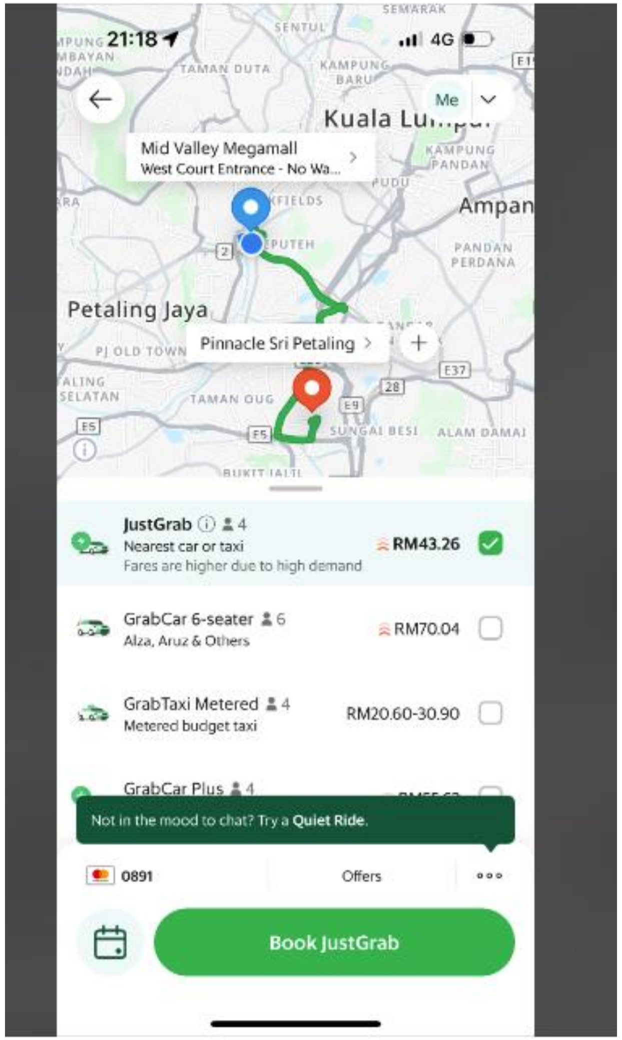
- Malaysia Prayer Times (by Norzaini Ilhami)
  - Uses Location-based service sensors to detect current city/town and display current prayer times.
  - Show Qibla direction based on users-location
    https://play.google.com/store/apps/details?id=com.i906.mpt

Example of Location Awareness Application

- **Grab**

Functionality: Grab is a super app offering ride-hailing, food delivery, and digital payment services. It uses GPS for real-time tracking of rides, food delivery, and determining fare or delivery costs based on the user's location.

Use Case: Users can book rides, order food, or even pay for goods and services through the app, with each service optimized based on the user's real-time location.



Context Awareness & Location Awareness

- Location Awareness is a **subset of Context-Awareness**
- Context Awareness app is able to provide based on additional data it gather from the users
- Some Location-Awareness app is also a Context Aware application

Context Awareness

"Context encompasses more than just the user's location, because other things of interest are also mobile and changing"

"People's actions can often be predicted by their situation. There are certain things we regularly do when in the library, kitchen or office. Contextual information aim to exploit this fact"

- B. Schilit; N. Adams; R. Want. (1994). "Context-aware computing applications". *IEEE Workshop on Mobile Computing Systems and Applications (WMCSA'94), Santa Cruz, CA, US*. pp. 89–10

Characteristics of Context Awareness Application

- Context Awareness application is able to **provide extra services or analysis based on additional data it gather from the users**
- Context Awareness application obtains its data from various mobile device **sensors**, user behavior **history** and **3rd party data sources**

Example of Context Awareness

EventBrite

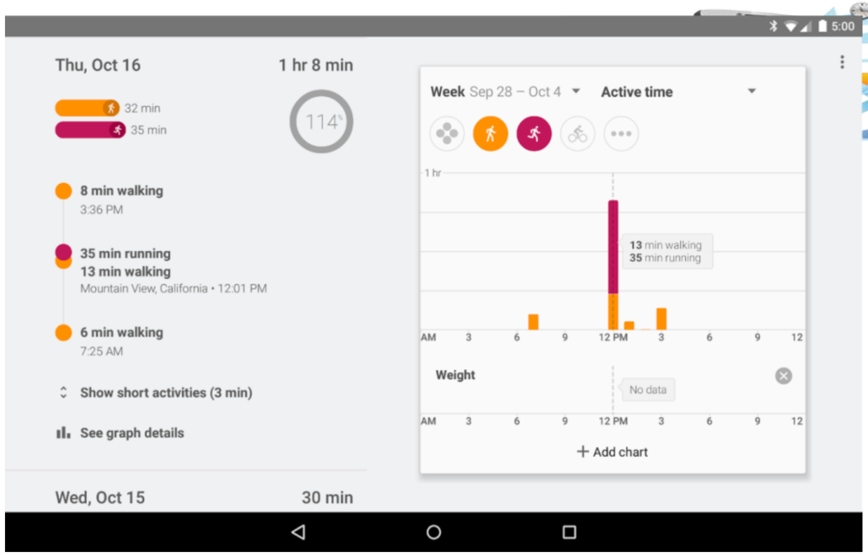
- In addition to detecting users location EventBrite also:
  - Scans device music folder to determine users' song preference
  - Save events that users' frequently viewed or attended
  - Scans Friends events
  - Determine your style and preference by analyzing users profile
    - Age, Horoscope signs, gender, income level, etc
- This are being done in order to present the most relevant information to users.

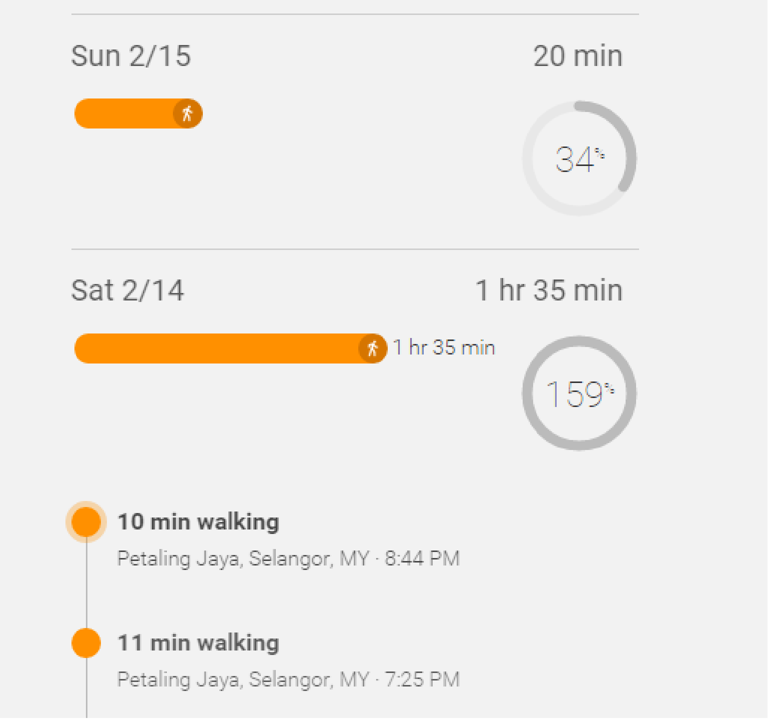
Example of Context Awareness

- **Google Fit**

In addition to detecting users location Google Fit also:

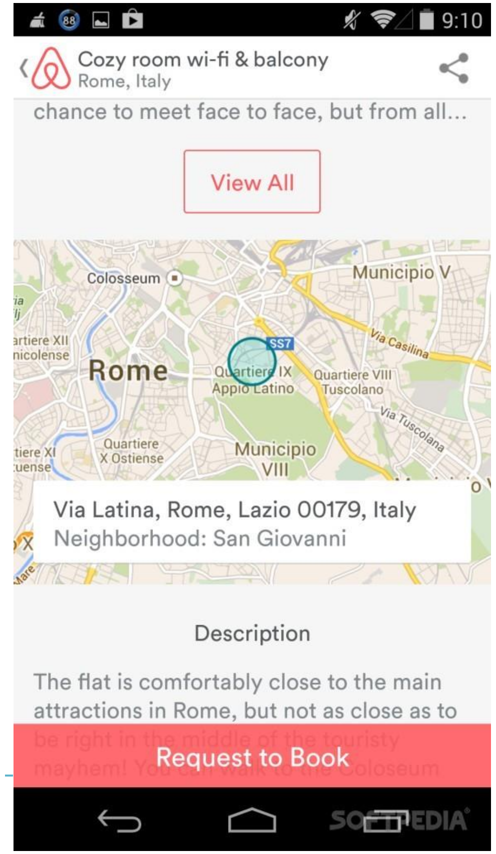
- Detects user activity based on accelerometer and gyroscope sensor reading
- Uses Pedometer sensor to detect users running , walking and cycling activity
- If connected to a smart watch, Google Fit can also measures heart rate and give advice to users on their fitness level





Example of Context Awareness

- **AirBnB**
  - Also known as **Air Bed and Breakfast**
  - An application which uses the users current location to list offers on private room to rent (or entire house) to rent
  - **AirBnB** also analyzes user profile based on previous behaviour to give better recommendation suited for the user



Context-Aware Applications

FASHION

stichfix

MUSIC

spotify
player.fm

FOOD

door dash
open table
yelp

MAPS + NAV

waze
google maps

SOCIAL

facebook
pinterest
instagram

ONLINE DATING

tinder
match

SHOPPING

amazon

TAXI

uber
lyft
sidecar

Location Aware vs Context Aware

Location-Aware

1. Detects user location
2. Focus on user location
3. Attempt to provides services or functionality based on user location
4. Use location data as input for application
5. Subset of Context-Aware
6. Where?

Context-Aware

1. May detect user location
2. Attempt to predict user behavior and provide suggestion based on user's activity
3. May access user profile, history or social-graph in order to determine context or connections
4. May access various mobile devices sensor to sense context
5. May involve data analysis in order to predict user behavior or context
6. + What, Who, How, When?

Privacy Issues of Location Awareness and Context Awareness Application

All types of context-aware applications must obtain user data in order to perform data analysis and provide suggestions.

The question of which type of data is recorded?

- Is the application is constantly monitoring for input?
- Does the application constantly record user location
- Does the back-end can deduce the location of user's work, school and home location ?
- This represent privacy issues.

LECTURE ENDS